Project Overview & Goals

The Python package `fleet_management` schedules a fleet of F trains over a horizon of T days. Every day, M missions must be flown. Each train flies at most one mission, spends the day in the depot, or stays idle. Each train has L components, and each component degrades stochastically. On a depot day a component may be repaired (imperfect repair) or replaced. The schedule is a mixed-integer program solved with Gurobi.

Goal. Minimise the cost of depot days, repairs and replacements, plus a penalty on the largest train damage (Section 4), subject to $\Pr(D_{i\ell k} > \tau_{i\ell}) \leq \varepsilon_{i\ell}$ for every component and every day, with $D_{i\ell k}$ the degradation and $\tau_{i\ell}$ the failure threshold. The transitory phase (H_1 days) starts from the current state of the fleet; the operating phase (H_2 days) is the schedule to repeat. The requirement holds on every day of the horizon and of every later repetition of the operating phase (Section 5.9).

Degradation models (input `model`, one per cell):

- **Gaussian**: Gaussian random walk; tracks mean and variance (Section **??**).
- **Inverse Gaussian (IG)**: tracks the mean; cap on the mean (Section **??**).
- **Wiener**: Brownian motion with drift; tracks mean and variance; the variance grows by σ^2 per mission day and falls only at a replacement (Section **??**).
- **Gamma**: Gamma increments with mission-dependent rates; tracks the mean; cap on the mean (Section **??**).
- **Rainflow**: distribution-free; the increment law enters through summary statistics; one of five tail bounds: Markov, Cantelli, Hoeffding, Bernstein, Chernoff (Section **??**).

Repair and replacement. Each cell has a repair law (Section 5.3): ARD_∞ (arithmetic reduction of deterioration with infinite memory, the default) or ARD_1 (arithmetic reduction of deterioration of order 1). Every model accepts both laws. Each cell may be replaced (Section 5.5). The loop row forces at least one replacement in the operating phase for a Wiener cell whose train flies a mission in that phase, and for an ARD_1 cell with $\rho_{i\ell} < 1$ that gains mean damage in that phase (scoping document, section "Stochastic loop constraint").

Problem class. A mixed-integer linear program (MILP) when every reliability row is linear and the damage penalty is `inf_norm` or absent ($C_D = 0$); otherwise a mixed-integer quadratically constrained program (MIQCP). It is nonconvex when a quadratic reliability row uses the form `exact` (Section 5.6), and convex when the only quadratic row is the `quadratic` damage penalty (Section 5.8).

Scope of the package. One solve (`solve`); horizon sweeps (Section **??**); warm starts (Section **??**); the schedule plot (Section **??**); validation tools (Section **??**). Inputs in YAML, JSON or HDF5 (Section **??**); outputs in Section **??**.

Authority. This specification is authoritative for the formulation and for the tools. Every change is written here first; the code follows it.

Non-goals.

- Routing decisions are not optimised: the missions are inputs.
- Continuous-time formulations are out of scope: the time step is one day.

Index Sets and Notation

Symbol	Meaning	Defined in
Indices and horizon		

Symbol	Meaning	Defined in
$i \in \{1, \dots, F\}$	train	
$\ell \in \{1, \dots, L\}$	component of a train	
(i, ℓ)	a *cell*: component ℓ of train i	
$j \in \{0, 1, \dots, M\}$	activity: $j = 0$ is a depot day, $j \geq 1$ a mission	
$k \in \{1, \dots, T\}$	day; index $k = 0$ holds the initial state	
$H = [H_1, H_2]$, $T = H_1 + H_2$	horizon input; a scalar H means $H_1 = H_2 = H$	??
days $1, \dots, H_1$	transitory phase	
days $H_1 + 1, \dots, T$	operating phase	5.9
$p \in \{1, \dots, P\}$	segment of the `pwl` form, $P =$ `pwl_points`	5.6
Parameters, increments and states		
$\tau_{i\ell}$, $\varepsilon_{i\ell}$, $\rho_{i\ell}$	failure threshold, reliability level, repair efficiency of a cell; a model section may drop the subscript $i\ell$ after saying so	??
C_M; $C_{R,i\ell}$, $C_{\text{rep},i\ell}$; C_D	costs: depot day; repair, replacement; damage penalty	4
$\mu^{\text{inc}}_{i\ell jk}$, $v^{\text{inc}}_{i\ell jk}$, $b^{\text{inc}}_{i\ell jk}$, $K^{\text{inc}}_{i\ell jk}$, $\beta_{i\ell jk}$	mean, variance, support width and cumulant generating function (CGF) of the increment of cell (i, ℓ) when train i flies mission j on day k, after the phase lookup; Gamma rate	??
$\delta\mu_{i\ell k} = \sum_{j=1}^{M} x_{ijk}\, \mu^{\text{inc}}_{i\ell jk}$	day increment of the mean; δv, δR (from $(b^{\text{inc}})^2$) and δK likewise; 0 on depot and idle days	
$\mu_{i\ell k}$, $v_{i\ell k}$, $R_{i\ell k}$, $K_{i\ell k}$	tracked mean, variance, Hoeffding budget, Chernoff quantity	3
$g^{\mu}_{i\ell k}$, $g^{v}_{i\ell k}$, $g^{R}_{i\ell k}$, $g^{K}_{i\ell k}$	anchors (law ARD_1)	5.3
$\mu_{0,i\ell}$, $v_{0,i\ell}$, $R_{0,i\ell}$, $K_{0,i\ell}$	initial values	??
$\mu^{\text{new}}_{i\ell}$, $v^{\text{new}}_{i\ell}$, $R^{\text{new}}_{i\ell}$, $K^{\text{new}}_{i\ell}$	replacement values	5.5
$b^{0}_{i\ell}$, $b^{\text{new}}_{i\ell}$	support widths of the initial and replacement damage	??
ARD_∞, ARD_1; a; ψ	repair laws; repair factor; repair map	5.3
Derived constants		
M_s	natural range of a tracked quantity s (big-M constant)	5.4
Φ	standard normal distribution function	??
$\kappa_G = \Phi^{-1}(1 - \varepsilon)$	standard normal quantile	??
$V_{\max} = \tau^2/\kappa_G^2$	variance range, Gaussian and Wiener	??
$\eta_{i\ell}$; $Z \sim \text{IG}(1, 1/\eta)$; q; $\bar{\mu} = \tau/\mathbb{E}[Z \mid Z > q]$	inverse Gaussian dispersion; unit law; its $(1 - \varepsilon)$-quantile; mean cap	??
$\beta^*_{i\ell}$; $t = \beta^*\tau$; $A_1(t, \varepsilon)$; $Q(a, x)$	smallest Gamma rate among the terms of the cell with a positive mean; scaled threshold; shape cap; regularised upper incomplete gamma function	??
$\kappa_C = \sqrt{(1 - \varepsilon)/\varepsilon}$	Cantelli coefficient	??
$\kappa = \ln(1/\varepsilon)$	exponent constant (Hoeffding, Bernstein)	??
$b_{i\ell}$, $c_{i\ell} = \kappa\, b_{i\ell}/3$	Bernstein support constant and margin	??
$s_{i\ell}$	Chernoff parameter (input `s_chernoff`)	??
V^{C}, V^{B}	variance ranges, rainflow Cantelli and Bernstein	??
$\mu_r =$ `tangent_ref` $\cdot\, \tau$; c_2, c_1	tangent point; constants of a quadratic row	5.6
E	evaluation horizon (input `evaluation_horizon`)	4

Decision Variables

A variable and its rows are built only where a row or an objective term with a non-zero coefficient uses them. The day index runs over $k = 1, \dots, T$; index $k = 0$ holds constants (initial values and starting anchors).

Variable	Shape, type	Built when	Meaning
x_{ijk}	$F \times (M{+}1) \times T$, binary	always	1 if train i flies mission j on day k; $j = 0$: depot day
$x^m_{i\ell k}$	$F \times L \times T$, binary	always	1 if component ℓ of train i is repaired on day k
$x^r_{i\ell k}$	$F \times L \times T$, binary	`allow_replacement = true`	1 if component ℓ of train i is replaced on day k
$n_{i\ell k}$	binary	cells with `encoding = "indicator"`	1 on a day without intervention, Eq. (15)
$\mu_{i\ell k}$	continuous, $[0, \tau_{i\ell}]$	every cell	tracked mean degradation
$v_{i\ell k}$	continuous, $[0, M_v]$	cells whose model or bound tracks a variance	tracked variance
$R_{i\ell k}$	continuous, $[0, M_R]$	rainflow cells with the Hoeffding bound	tracked Hoeffding budget
$K_{i\ell k}$	continuous, $[0, M_K]$	rainflow cells with the Chernoff bound	tracked Chernoff quantity
$g^{\mu}_{i\ell k}$, $g^{v}_{i\ell k}$, $g^{R}_{i\ell k}$, $g^{K}_{i\ell k}$	continuous, range of the quantity	ARD_1 cells, one per tracked quantity except the Wiener variance	anchors, Eq. (10)
$w_{i\ell kp}$	binary, $p = 1, \ldots, P$	cells with a quadratic row in the `pwl` form, $P \geq 2$	segment of the mean, Eq. (20)
$z_{i\ell k}$	continuous, ≥ 0	cells with $C_{R,i\ell} > 0$, on days whose cost enters the objective	degradation removed by a repair, Section 5.7
u	scalar, ≥ 0	$C_D > 0$	damage regularisation, Section 5.8

M_v, M_R, M_K are the natural ranges of Section 5.4. The output reports the tracked values; each is an upper bound on the value that the rules give with every row at equality (scoping document, section "Stochastic loop constraint").

Objective Function

Cost of day k, and costs of the two phases:

$$c_k = C_M \sum_{i=1}^{F} x_{i0k} + \sum_{i=1}^{F} \sum_{\ell=1}^{L} C_{R,i\ell}\, z_{i\ell k} + \sum_{i=1}^{F} \sum_{\ell=1}^{L} C_{\mathrm{rep},i\ell}\, x^r_{i\ell k}, \tag{1}$$

$$J_{\mathrm{init}} = \sum_{k=1}^{H_1} c_k, \qquad J_{\mathrm{op}} = \sum_{k=H_1+1}^{T} c_k. \tag{2}$$

$z_{i\ell k}$ reads 0 where it is not built; the last sum of c_k is absent when `allow_replacement = false`. The input `objective_mode` (default `total`) selects the objective:

$$\min \begin{cases} J_{\mathrm{init}} + J_{\mathrm{op}} + C_D\, u & \texttt{total}, \\ J_{\mathrm{op}}/H_2 + C_D\, u & \texttt{operating_average}, \\ J_{\mathrm{init}} + \dfrac{E - H_1}{H_2}\, J_{\mathrm{op}} + C_D\, u & \texttt{evaluation_total}, \end{cases} \tag{3}$$

with E the input `evaluation_horizon`, required in the mode `evaluation_total`, $E > H_1$. The term $C_D\, u$ is present only when $C_D > 0$.

Cost terms (shapes in Section **??**).

- $C_M \geq 0$, scalar: paid once per train per depot day ($x_{i0k} = 1$), whatever the number of interventions on that day. An idle day costs nothing.
- $C_{R,i\ell} \geq 0$: per unit of mean degradation removed by a repair (Section 5.7).
- $C_{\text{rep},i\ell} \geq 0$: per replacement; required, no default.
- $C_D \geq 0$: weight of u (Section 5.8); the term is added once in every mode.

Constraints

Every row holds for every day $k = 1, \ldots, T$ unless its section says otherwise.

Assignment Constraints

Each train performs at most one activity per day:

$$\sum_{j=0}^{M} x_{ijk} \leq 1 \qquad \forall\, i, k. \tag{4}$$

Each mission is flown by exactly one train every day:

$$\sum_{i=1}^{F} x_{ijk} = 1 \qquad \forall\, j \in \{1, \ldots, M\},\ k. \tag{5}$$

Exactly M trains fly every day; the other $F - M$ trains are in the depot or idle. No other fleet-availability row is built. An idle day is an all-zero assignment row. Depot days and idle days add no increment.

Depot-Day Compatibility

A component may be repaired or replaced only on a depot day of its train, and at most one of the two on the same day:

$$x^m_{i\ell k} + x^r_{i\ell k} \leq x_{i0k} \qquad \forall\, i, \ell, k. \tag{6}$$

With `allow_replacement = false` the row reads $x^m_{i\ell k} \leq x_{i0k}$. Several components of one train may receive interventions on the same depot day; a component without an intervention keeps its state.

Repair Laws

The input `repair_model` selects the law per cell (default `ardinf`); the repair efficiency is $\rho = \rho_{i\ell} \in (0, 1]$. With D^- the degradation before a repair on day k (end of day $k-1$) and D^+ the degradation after it:

$$\text{ARD}_\infty \text{ (`ardinf`)}: \quad D^+ = (1 - \rho)\, D^-, \tag{7}$$

$$\text{ARD}_1 \text{ (`ard1`)}: \quad D^+ = D^- - \rho \left(D^- - D^{\text{last}}\right), \tag{8}$$

with D^{last} the level just after the last intervention (repair or replacement); before the first intervention of the horizon, the level just after the last intervention before it. Every model accepts both laws.

Repair map. For a tracked quantity s with anchor g, a repair on day k gives

$$\psi(s_{k-1}, g_{k-1}) = \begin{cases} a\, s_{k-1} & \text{ARD}_\infty, \\ a\, s_{k-1} + (1 - a)\, g_{k-1} & \text{ARD}_1, \end{cases} \tag{9}$$

with the repair factor $a = 1 - \rho$ for a mean and for K, $a = (1-\rho)^2$ for a variance and for R, and $a = 1$ for the Wiener variance (both laws). The model sections give the factor of each tracked quantity.

Anchors (ARD_1 only; one per tracked quantity, none for the Wiener variance):

$$g_0 = \text{starting anchor}, \qquad g_k = \begin{cases} s_k & \text{on a repair or replacement day,} \\ g_{k-1} & \text{otherwise.} \end{cases} \tag{10}$$

The starting anchors $g^{\mu}_{i\ell 0}$, $g^{v}_{i\ell 0}$, $g^{R}_{i\ell 0}$, $g^{K}_{i\ell 0}$ are the optional inputs `anchor_mu_0`, `anchor_v_0`, `anchor_R_0`, `anchor_K_0`; the default of each is the initial value of its quantity; each satisfies $0 \le g_0 \le s_0$ (Section **??**).

Encoding of the Dynamics

Every dynamics row is one-sided ($\ge$); replacement rows and anchor rows are equalities. For one tracked quantity s of a cell (a mean, a variance, R or K): day increment δs_k, natural range M_s, replacement value s^{new}, anchor g (ARD_1 only), repair map ψ (Eq. (9)), cell binaries x^m_k, x^r_k, and $e_k = x^m_k + x^r_k$. The input `encoding` selects the form per cell.

Big-M form (`encoding = "bigm"`, the default):

$$\text{no intervention:} \quad s_k \ge s_{k-1} + \delta s_k - M_s\,(x^m_k + x^r_k), \tag{11}$$

$$\text{repair:} \quad s_k \ge \psi(s_{k-1}, g_{k-1}) - M_s\,(1 - x^m_k), \tag{12}$$

$$\text{replacement:} \quad |s_k - s^{\mathrm{new}}| \le M_s\,(1 - x^r_k), \tag{13}$$

$$\text{anchor } (\mathrm{ARD}_1): \quad |g_k - s_k| \le M_s\,(1 - e_k), \qquad |g_k - g_{k-1}| \le M_s\,e_k. \tag{14}$$

Each row with an absolute value is built as two linear rows.

Indicator form (`encoding = "indicator"`). One binary $n_{i\ell k}$ per cell and day, and Gurobi indicator constraints:

$$n_{i\ell k} + x^m_{i\ell k} + x^r_{i\ell k} = 1, \tag{15}$$

$$n_k = 1 \;\Rightarrow\; s_k \ge s_{k-1} + \delta s_k, \quad x^m_k = 1 \;\Rightarrow\; s_k \ge \psi(s_{k-1}, g_{k-1}), \quad x^r_k = 1 \;\Rightarrow\; s_k = s^{\mathrm{new}}, \tag{16}$$

$$n_k = 1 \;\Rightarrow\; g_k = g_{k-1}, \quad x^m_k = 1 \;\Rightarrow\; g_k = s_k, \quad x^r_k = 1 \;\Rightarrow\; g_k = s_k. \tag{17}$$

Rules for both forms.

- Anchor rows only under ARD_1, for every tracked quantity except the Wiener variance.
- Wiener variance: $\psi = s_{k-1}$, no anchor; it is never left free below on a repair day. In the big-M form its no-intervention and repair rows may be replaced by the single row Eq. (**??**).
- `allow_replacement = false`: no x^r; read $x^r_k = 0$ and drop the replacement rows.
- At binary points the two forms have the same feasible set.

Natural ranges M_s (big-M constants): $\tau_{i\ell}$ for a mean; $V_{\max}$ for a Gaussian or Wiener variance; V^{C} or V^{B} for a rainflow variance (Cantelli or Bernstein); $2\tau^2/\kappa$ for R; $s_{i\ell}\tau - \kappa$ for K (Section **??**). An anchor has the range of its quantity.

Replacement

The input `allow_replacement` (default `true`) makes replacements available; when `false`, the binaries $x^r_{i\ell k}$, the replacement rows and the replacement cost are not built. On a replacement day every tracked quantity of the cell takes its replacement value, and every anchor the same value:

- $\mu^{\mathrm{new}}_{i\ell}$: input `replacement_mu`, default 0;

- $v_{i\ell}^{\text{new}}$: input `replacement_v`, default 0, where a variance is tracked;
- $R_{i\ell}^{\text{new}}$, $K_{i\ell}^{\text{new}}$: rainflow Hoeffding and Chernoff, Section **??**.

The rows are Eq. (13) or Eq. (16). Each replacement value lies in the natural range of its quantity, and the replacement state satisfies every row of a replacement day in the form built at the cell (Section **??**).

The input `replacement_as_new` (default `false`) sets every replacement value of the cell to 0; a non-zero replacement input together with it is an input error. A replacement costs $C_{\text{rep},i\ell}$ only ($z_{i\ell k} = 0$ on a replacement day). A cell may be replaced on any depot day, any number of times.

Linear Forms of the Reliability Rows

The Gaussian, Wiener, Cantelli, Hoeffding and Bernstein rows have the form

$$\chi_{i\ell k} \;\le\; f(\mu_{i\ell k}), \qquad f(\mu) = c_2\,(\tau-\mu)^2 + c_1\,(\tau-\mu), \qquad \mu_{i\ell k} \le \bar{\mu}_{\text{side}}, \tag{18}$$

with $f'(\mu) = -2c_2(\tau-\mu) - c_1$; the third relation is the side row.

Row	χ	c_2	c_1	$\bar{\mu}_{\text{side}}$
Gaussian, Wiener	v	$1/\kappa_G^2$	0	τ
Rainflow, Cantelli	v	$1/\kappa_C^2 = \varepsilon/(1-\varepsilon)$	0	τ
Rainflow, Hoeffding	R	$2/\kappa$	0	τ
Rainflow, Bernstein	v	$1/(2\kappa)$	$-b_{i\ell}/3$	$\tau - c_{i\ell}$

The input `reliability_impl` selects the form per cell; every form keeps the side row.

- `exact`: Eq. (18), a nonconvex quadratic constraint. Gurobi's `NonConvex` is set to 2 when at least one cell has a quadratic row in this form.
- `tangent`: one linear row per day, at $\mu_r = \texttt{tangent_ref}_{i\ell} \cdot \tau_{i\ell}$ (`tangent_ref` per cell, in $[0,1)$, default 0.5):

  $$\chi_{i\ell k} \;\le\; f(\mu_r) + f'(\mu_r)\,(\mu_{i\ell k} - \mu_r). \tag{19}$$

  At a Bernstein cell, $\mu_r < \sqrt{\tau\,(\tau - 2c_{i\ell})}$ (Section **??**).
- `pwl`: $P = $ `pwl_points` equal segments $[(p-1)\tau/P,\ p\tau/P]$ of $[0,\tau]$ (default $P = 8$), midpoints $\hat{\mu}_p = (p-\frac{1}{2})\tau/P$, and one binary $w_{i\ell kp}$ per cell, day and segment:

  $$\textstyle\sum_{p=1}^{P} w_{i\ell kp} = 1, \tag{20}$$

  $$w_{i\ell kp} = 1 \;\Rightarrow\; (p-1)\,\tau/P \le \mu_{i\ell k} \le p\,\tau/P, \tag{21}$$

  $$w_{i\ell kp} = 1 \;\Rightarrow\; \chi_{i\ell k} \le f(\hat{\mu}_p) + f'(\hat{\mu}_p)\,(\mu_{i\ell k} - \hat{\mu}_p). \tag{22}$$

  The implications are Gurobi indicator constraints in both encodings. With $P = 1$: the single tangent at $\tau/2$, no binaries. At a Bernstein cell, $\tau/(2P) < \sqrt{\tau\,(\tau - 2c_{i\ell})}$ (Section **??**).

At a cell whose reliability row is linear (Markov, Chernoff, inverse Gaussian, Gamma) the row is built in its exact form and the value of `reliability_impl` is ignored. An unknown value is an input error at every cell. Each linear form implies the exact row (scoping document, section "Conservative surrogates for the empirical damage-increment distribution").

Repair Cost Variable

$z_{i\ell k} \ge 0$ is the mean degradation removed by a repair of cell (i,ℓ) on day k, for every model and both laws. It is built at cells with $C_{R,i\ell} > 0$, on every day $k = 1,\dots,T$ except the transitory days in the mode `operating_average`. Only the lower rows are built:

$$z_{i\ell k} \;\ge\; \rho_{i\ell}\left(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1}\right) - \rho_{i\ell}\,\tau_{i\ell}\left(1 - x^{m}_{i\ell k}\right), \qquad z_{i\ell k} \ge 0, \tag{23}$$

with $g^{\mu} \equiv 0$ under ARD_{∞}. At the optimum $z_{i\ell k} = 0$ on every day without a repair, replacement days included. z is an upper bound on the expected removed degradation, except at the first repair when the starting anchor is above the true one (scoping document, section "Cost function").

Damage Regularisation

The scalar $u \geq 0$ and its rows are built only when $C_D > 0$; with $C_D = 0$ the input `penalty_type` is ignored. One row per train and day:

$$\texttt{inf_norm}\ \text{(default)}: \quad u \;\geq\; \sum_{\ell=1}^{L} \mu_{i\ell k} \qquad \forall\, i, k, \tag{24}$$

$$\texttt{quadratic}: \quad u \;\geq\; \sum_{\ell=1}^{L} \mu_{i\ell k}^{2} \qquad \forall\, i, k. \tag{25}$$

At the optimum $u = \max_{i,k} \sum_{\ell} \mu_{i\ell k}$ (`inf_norm`) or $u = \max_{i,k} \sum_{\ell} \mu_{i\ell k}^{2}$ (`quadratic`): the largest damage of one train on one day. Eq. (25) is a rotated second-order cone; it does not need `NonConvex`. If $C_D > 0$ and `penalty_type = "quadratic"`, every cell with a quadratic reliability row must have `reliability_impl = "exact"`; otherwise a `ValueError` is raised (Section **??**). At a cell with a linear row the value is ignored. With $C_D = 0$ the rule does not apply.

Loop Constraint

One hard row per cell and per coordinate s of its tracked state:

$$s_{i\ell T} \;\leq\; s_{i\ell H_1}. \tag{26}$$

The rows cover every tracked quantity of the cell (the mean, and the variance, R or K where tracked) and, under ARD_1, every anchor. Each coordinate has its own row; no row compares a scalar function of them, such as a failure-probability bound. There is no soft loop penalty. With these rows every day of every repetition of the operating phase satisfies the reliability requirement, for every model, both laws, both encodings, every form, and any H_1, H_2 (scoping document, section "Stochastic loop constraint").

Consequences. An ARD_1 cell with $\rho_{i\ell} < 1$ whose train flies, in the operating phase, a mission with a positive mean increment for the cell must be replaced at least once in that phase; with `allow_replacement = false` its train cannot fly such a mission there, and the model is infeasible if the fleet cannot cover the missions otherwise (scoping document, section "Stochastic loop constraint"). A Wiener cell likewise needs a replacement in the operating phase when its train flies a mission there (Section **??**).

References

References

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